Section A

1 (a)	work done moving <u>unit</u> mass	M1	
	from infinity to the point	A1	[2]
(b) (i)	at R, $\phi = 6.3 \times 10^{7}$ J kg^{-1} (allow $\pm\, 0.1 \times 10^{7}$)	B1	
	$\phi = GM / R$		
	$6.3 \times 10^{7} = (6.67 \times 10^{-11} \times M) / (6.4 \times 10^{6})$	C1	
	$M = 6.0 \times 10^{24}$ kg (allow $5.95 \rightarrow 6.14$)	A1	[3]
	Maximum of 2/3 for any value chosen for ϕ not at R		
(ii)	change in potential $= 2.1 \times 10^{7}$ J kg^{-1} (allow $\pm\, 0.1 \times 10^{7}$)	C1	
	loss in potential energy = gain in kinetic energy	B1	
	$\frac{1}{2} mv^{2} = \phi$m or $\frac{1}{2} mv^{2} = GM / 3R$	C1	
	$\frac{1}{2} v^{2} = 2.1 \times 10^{7}$		
	$v = 6.5 \times 10^{3}$ m s^{-1} ……………….(allow $6.3 \rightarrow 6.6$)	A1	[4]
	(answer 7.9×10^{3} m s^{-1}, based on $x = 2R$, allow max 3 marks)		
(iii)	e.g. speed / velocity / acceleration would be greater	B1	
	deviates / bends from straight path	B1	[2]
	(any sensible ideas, 1 each, max 2)		